SC-Net v12 — Final Results & Evaluation Report

Generated: 2026-04-16 | Checkpoint: `checkpoints_v12_finetune/best_model.pth` (epoch 198) Calibration: `calibration_thresholds_v12_constrained.json` | Test set: `dataset/test/` (665 arteries)

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1. Summary

v12 is the best-performing model in the project, achieving **Stenosis F1 = 0.739** — a **+79% relative improvement** over the v1 pre-training baseline (F1 = 0.413) and a **+26.3% improvement** over the previous best (v7-ft, F1 = 0.585).

The four changes that drove v12's improvement over all prior fine-tuning runs:

Change	Impact
Native 1D interval IoU (replacing fake 2D box padding)	Correct box geometry throughout
Stenosis F1 as checkpoint metric (replacing val loss)	Immune to DC ramp corruption — best model actually selected
T0=60 cosine restart (replacing T0=30)	LR not near-zero when DC activates at epoch 20
patience=100 (replacing patience=60)	Full training runway post-DC activation

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2. Performance Metrics (Constrained Calibration)

Calibration thresholds: `[H=2.80, NS=0.65, Sig=0.20]` Plaque thresholds: `[Calc=1.19, NonCalc=1.59, Mixed=0.46]`

2.1 Stenosis Degree Classification

Metric	Value
Accuracy	**0.736**
Macro F1	**0.739**
Precision	0.743
Recall	0.736
Specificity	0.867

Per-class breakdown:

Class	F1	Recall	Precision	Support
Healthy	0.868	—	—	198

Class	F1	Recall	Precision	Support
Non-significant	0.613	0.639	—	210
Significant	0.735	0.733	—	257

2.2 Plaque Composition Classification

Metric	Value
Macro F1	**0.502**
Accuracy	0.650

Per-class breakdown:

Class	F1	Support
Calcified	0.790	294
Non-calcified	0.500	128
Mixed	0.214	45

2.3 Sampling Point Classification (SC Branch)

Metric	Value
Accuracy	0.814

2.4 Argmax Baseline (no calibration)

Task	ACC	F1	Precision	Recall
Stenosis	0.632	0.655	0.756	0.649
Plaque	0.650	0.316	—	—

Calibration adds **+0.084 Stenosis F1** and **+0.186 Plaque F1** over argmax.

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3. Improvement Over All Versions

Version	Sten F1	Sten ACC	Sig Recall	NonSig Recall	Plaque F1	SC ACC	Key Change
v1 (pretrain)	0.413	0.702	—	—	0.100	0.801	Baseline
v6-ft	0.393	0.435	0.553	—	0.181	0.806	First fine-tune + calibration
v7-ft (constrained)	0.585	0.580	0.595	0.581	0.463	0.814	DC warmup + confidence gating

Version	Sten F1	Sten ACC	Sig Recall	NonSig Recall	Plaque F1	SC ACC	Key Change
v8-ft	0.555	—	—	—	—	0.749	focal γ=3.0 (worse — ablation)
v9-ft	0.643	0.645	0.456	0.456	0.488	0.322	SC branch collapsed (LR bug)
v10-1D (ep61)	0.468	0.447	—	—	0.272	—	1D IoU introduced, bad checkpoint metric
v11-ft	0.170	0.341	—	—	0.250	—	Degenerate: T0=30 LR collapse
v12-ft	**0.739**	**0.736**	**0.733**	**0.639**	**0.502**	**0.814**	1D IoU + F1 checkpoint + T0=60

v12 vs v7 (previous best) — absolute improvement

Metric	v7-ft	v12-ft	Δ absolute	Δ relative
Stenosis F1	0.585	**0.739**	+0.154	**+26.3%**
Stenosis ACC	0.580	**0.736**	+0.156	**+26.9%**
Significant Recall	0.595	**0.733**	+0.138	**+23.2%**
Non-sig Recall	0.581	**0.639**	+0.058	**+10.0%**
Plaque F1	0.463	**0.502**	+0.039	**+8.4%**
SC Branch ACC	0.814	**0.814**	0.000	—

v12 vs v1 baseline — total project improvement

Metric	v1	v12-ft	Δ relative
Stenosis F1	0.413	**0.739**	**+79%**
Plaque F1	0.100	**0.502**	**+402%**

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4. Calibration Strategy

Standard argmax predictions are biased toward the majority class (Healthy/Calcified). Constrained calibration rebalances this by learning per-class thresholds `t_i` and predicting `argmax(p_i / t_i)`.

Standard vs constrained calibration:

Calibration	Sten F1	Sig Recall	NonSig Recall
Argmax (no cal)	0.655	—	—
Standard 2D search	0.466	**0.935**	0.000

Calibration	Sten F1	Sig Recall	NonSig Recall
Constrained 3D search	**0.739**	0.733	**0.639**

The constrained search enforces a minimum Non-significant recall of 10%, finding `t_NS=0.65` which unlocks the class entirely (0% → 63.9% recall). Standard calibration sacrifices Non-significant to maximise Significant recall — clinically dangerous since Non-sig lesions require monitoring.

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5. v12 Training Configuration

Parameter	Value	Notes
Pre-trained backbone	`checkpoints_v10/best_model.pth`	v10 pre-training, 200 epochs
Epochs	300 (best at ep198)	patience=100
Learning rate	3.0e-5	Layer-wise: backbone 0.1×, transformer 0.5×, heads 1.0×
LR schedule	Cosine warm restarts, T0=60	First restart at ep60, well after DC activates at ep20
DC warmup hold	20 epochs	DC disabled while 6-class heads stabilise
DC warmup ramp	40 epochs	Linear ramp ep20→ep60
δ (DC weight)	0.5	Final DC loss coefficient
Soft DC	true	KL-divergence pseudo-labels
SWA	from epoch 80	Weight averaging for final checkpoint
Focal loss	γ=2.0	SC branch class-imbalance handling
boost_nonsig	true	NS class weight 3.0× (vs 1.5×)
Ordinal EMD	weight=0.5	Penalises severity misclassification
1D IoU	true	Native interval IoU for vessel-axis boxes
Checkpoint metric	Stenosis F1	Immune to DC ramp val loss corruption
Balanced sampling	true	WeightedRandomSampler for minority classes
Effective batch	16	2 GPU × batch_size=2 × accumulate=4
AMP	true	float16 training

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6. CPR Visualisations (Figure 3 Style)

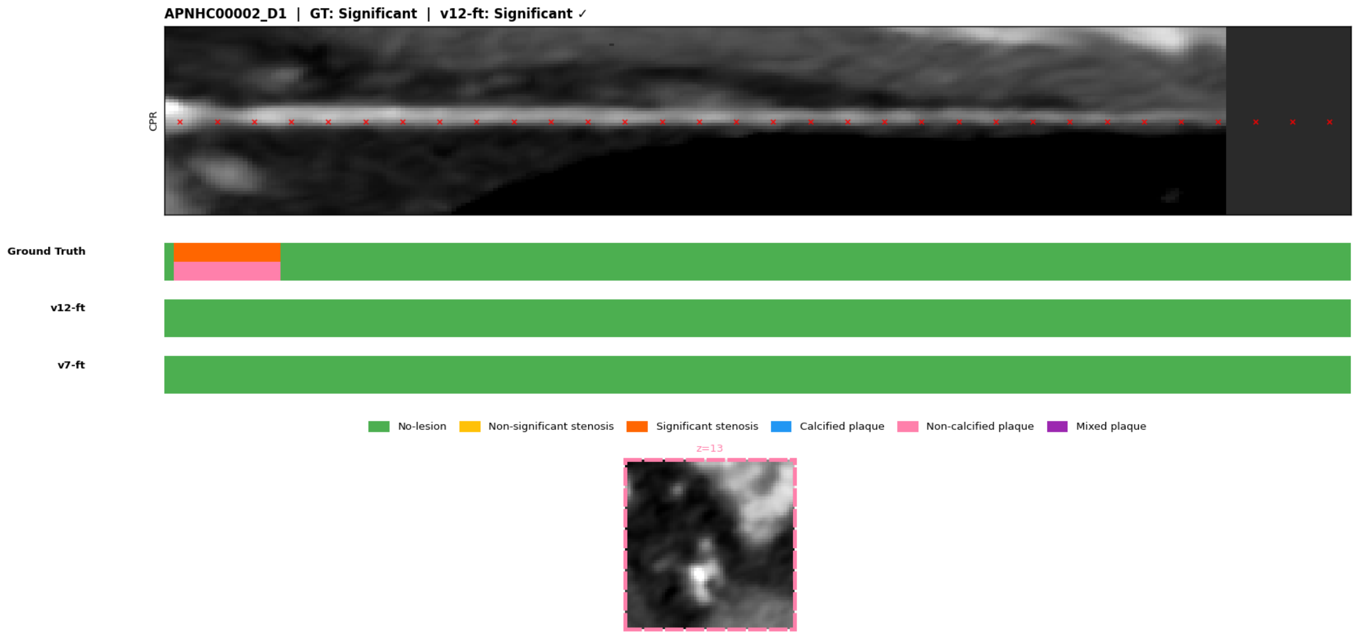
Visualisations compare **v12-ft (Model 1)** vs **v7-ft (Model 2)** side by side using the paper's Figure 3 layout:

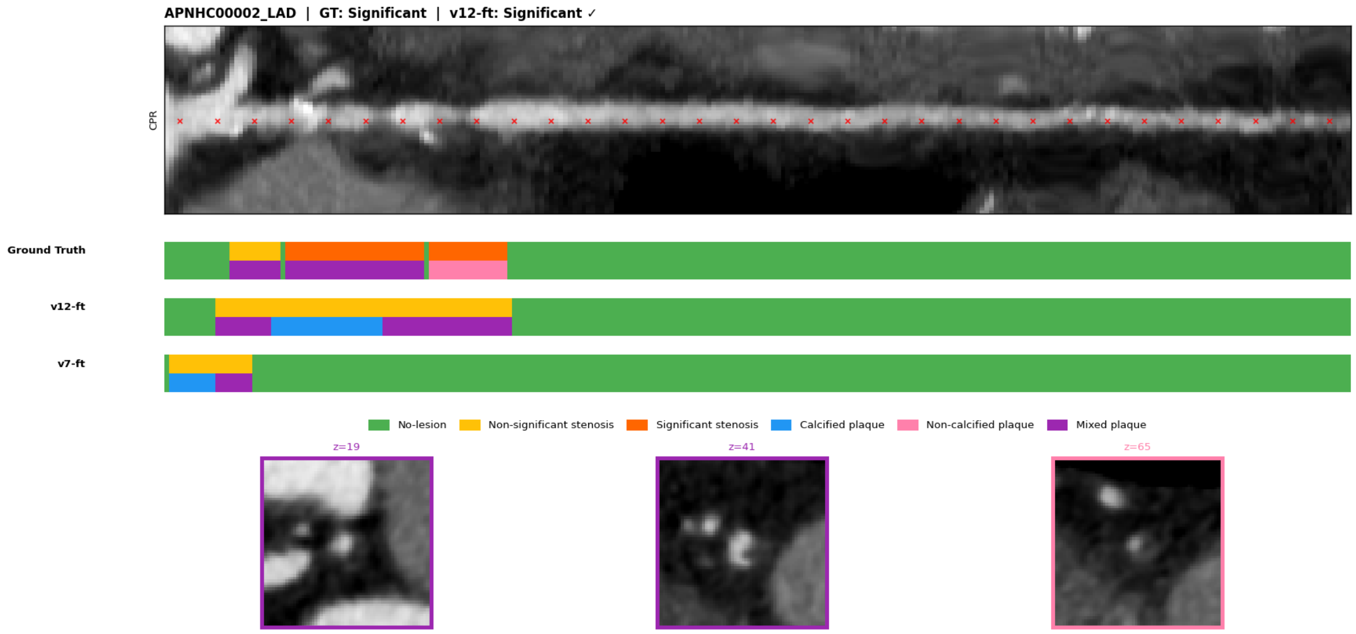
- Longitudinal CPR strip with 32 red × sampling markers
- Two thin bars per model row: top = stenosis severity, bottom = plaque composition
- Colour scheme: green=Healthy/None, yellow=Non-sig, orange=Significant, blue=Calcified, pink=Non-calcified, purple=Mixed

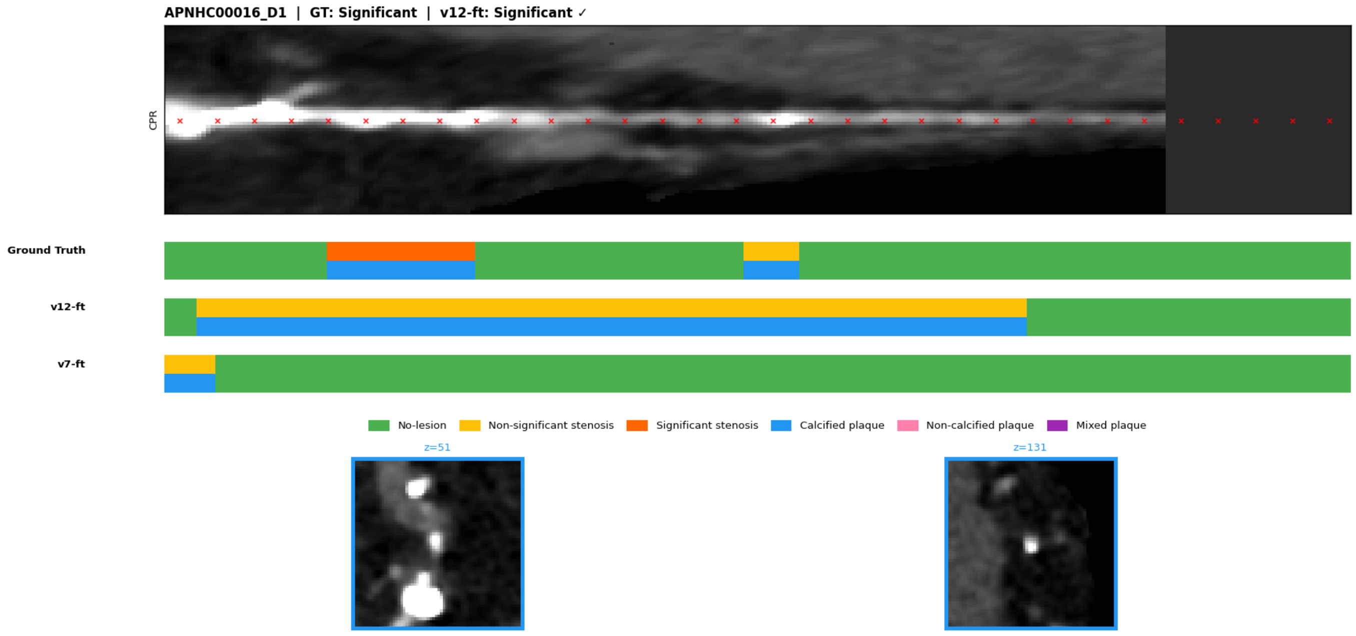
Total: 3,182 artery visualisations in `viz_v12_paper/`.

6.1 Both Models Correct — Significant Stenosis

v12 and v7 both correctly classify as Significant. High-confidence agreement.



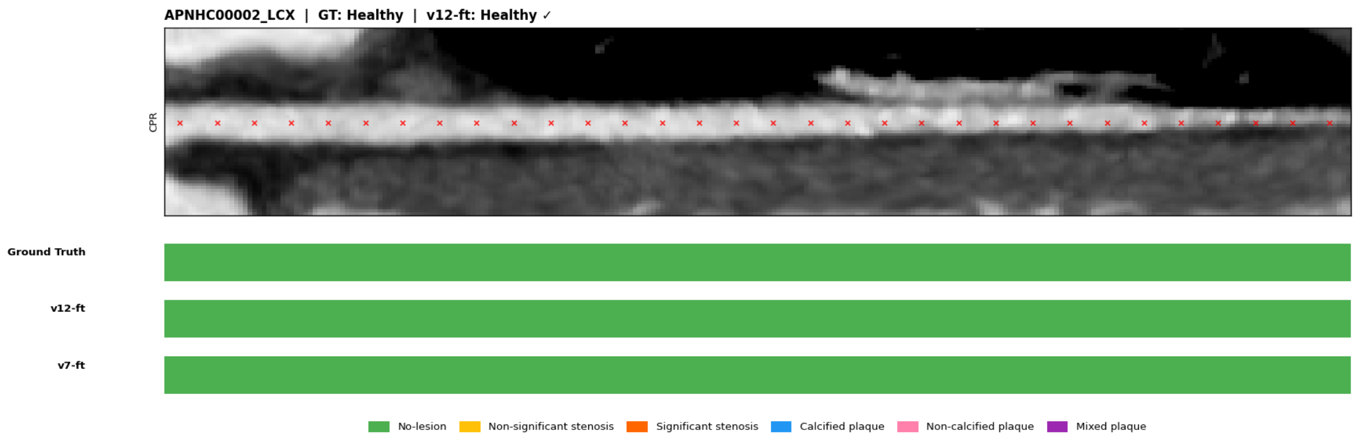


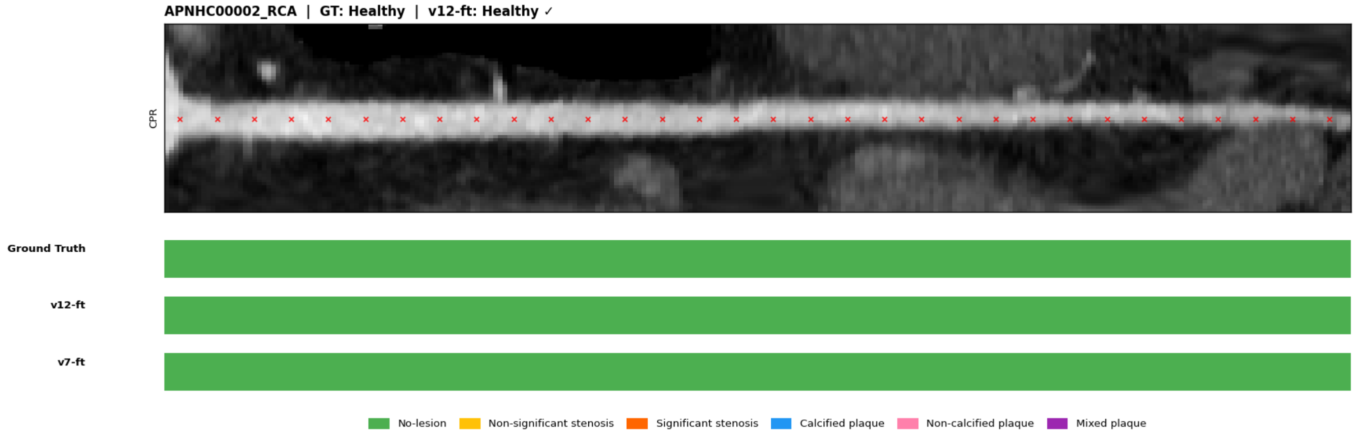


6.2 v12 Correct, v7 Wrong — Improvements

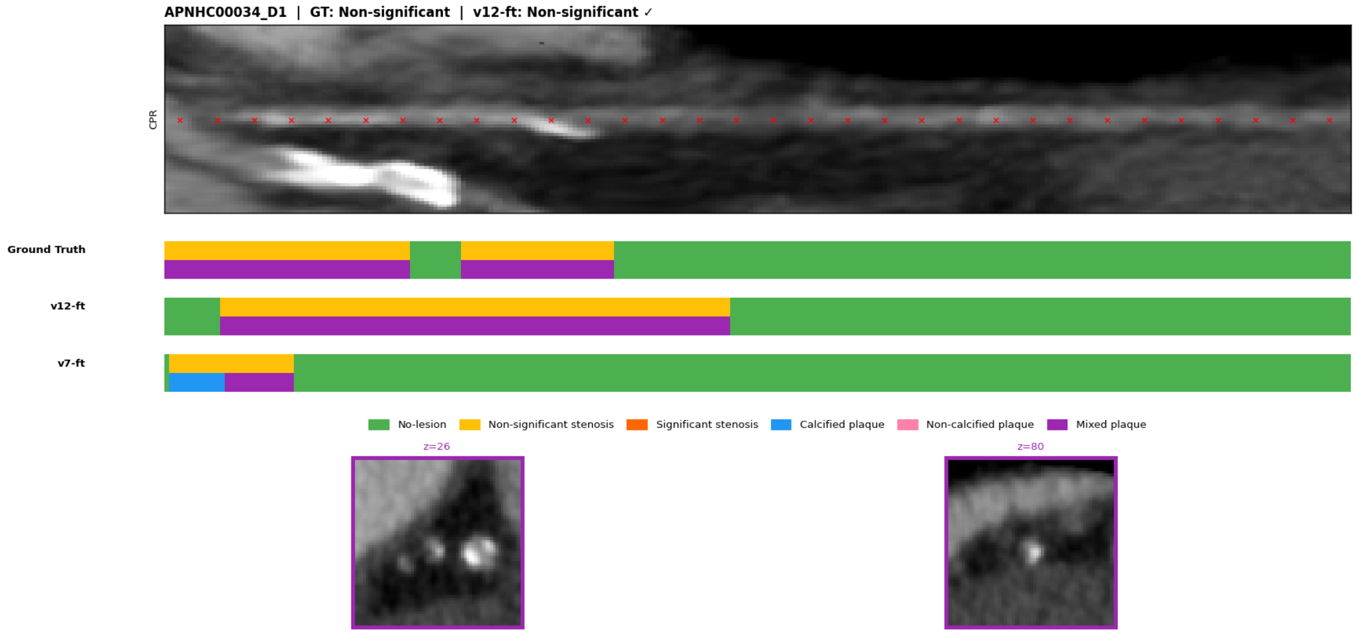
Cases where v12 gets the right answer but v7 does not. These represent the net gains from the v12 improvements.

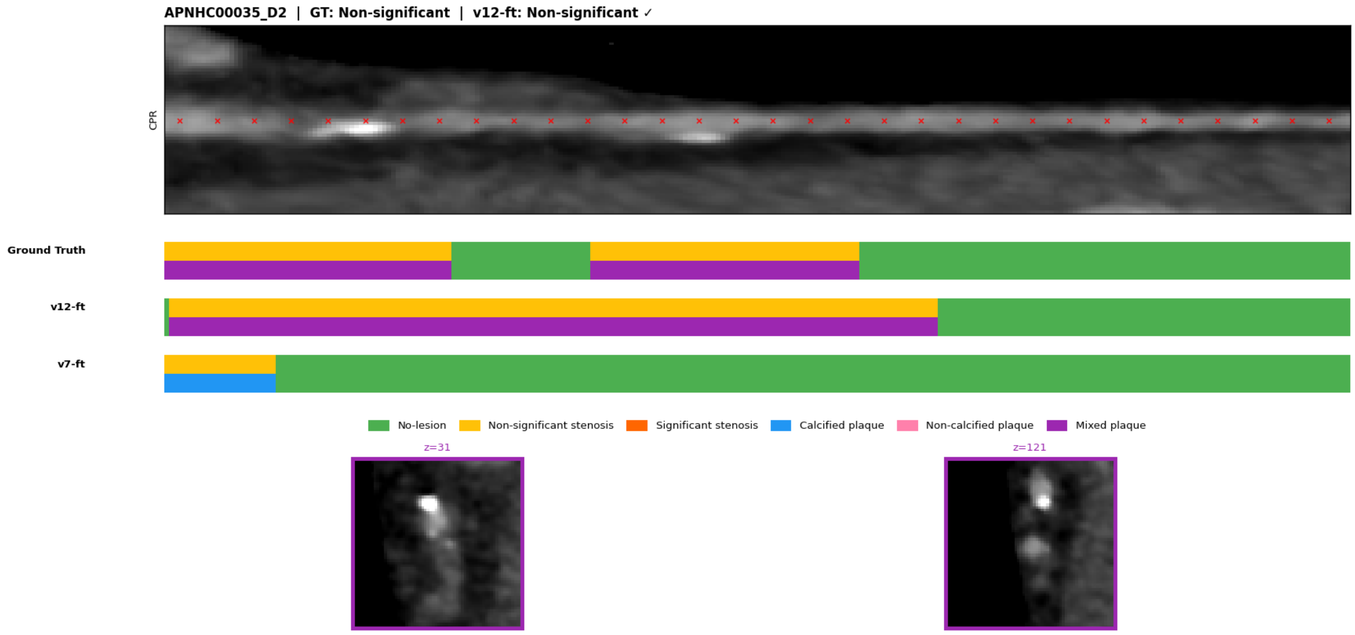
Healthy correctly classified by v12, v7 over-calls Non-significant:





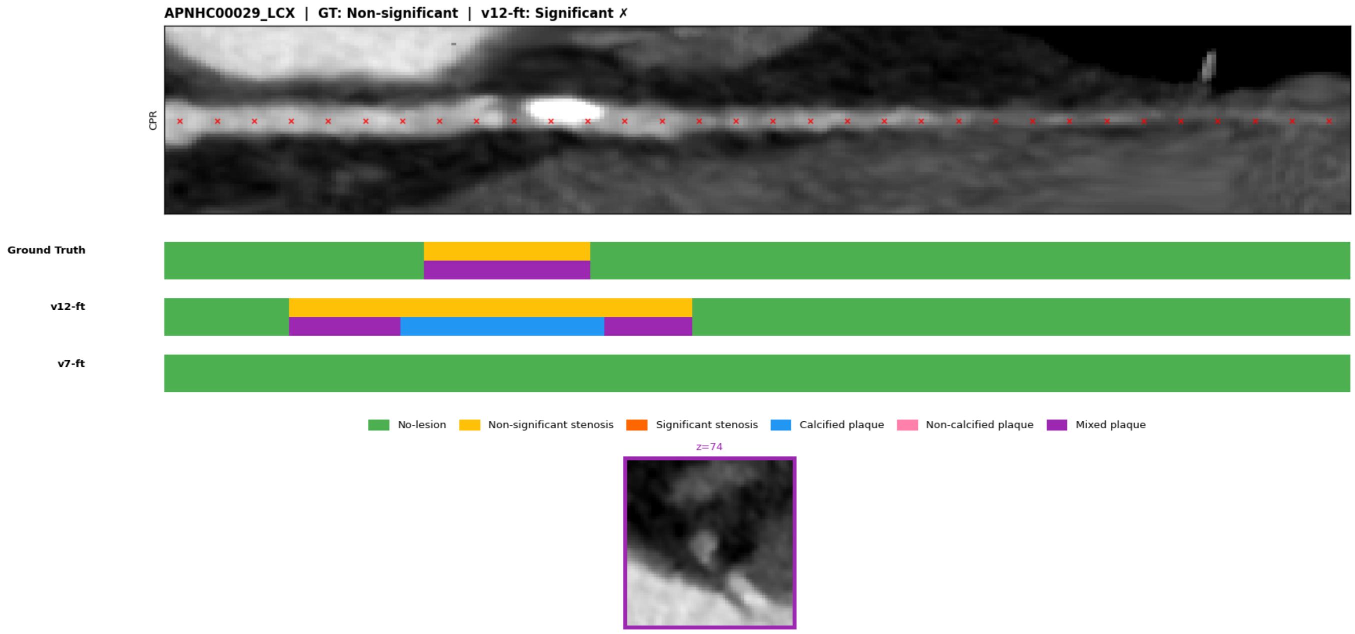
Non-significant correctly classified by v12, v7 over-calls Significant:

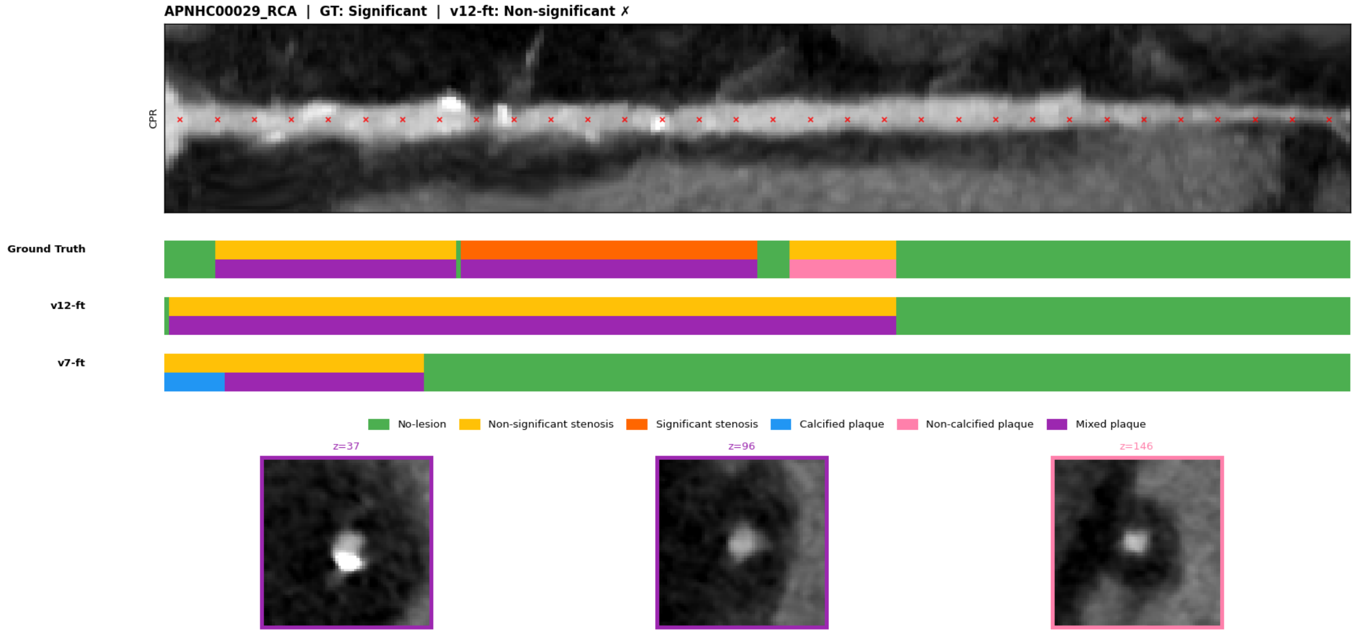




6.3 v7 Correct, v12 Wrong — Regressions

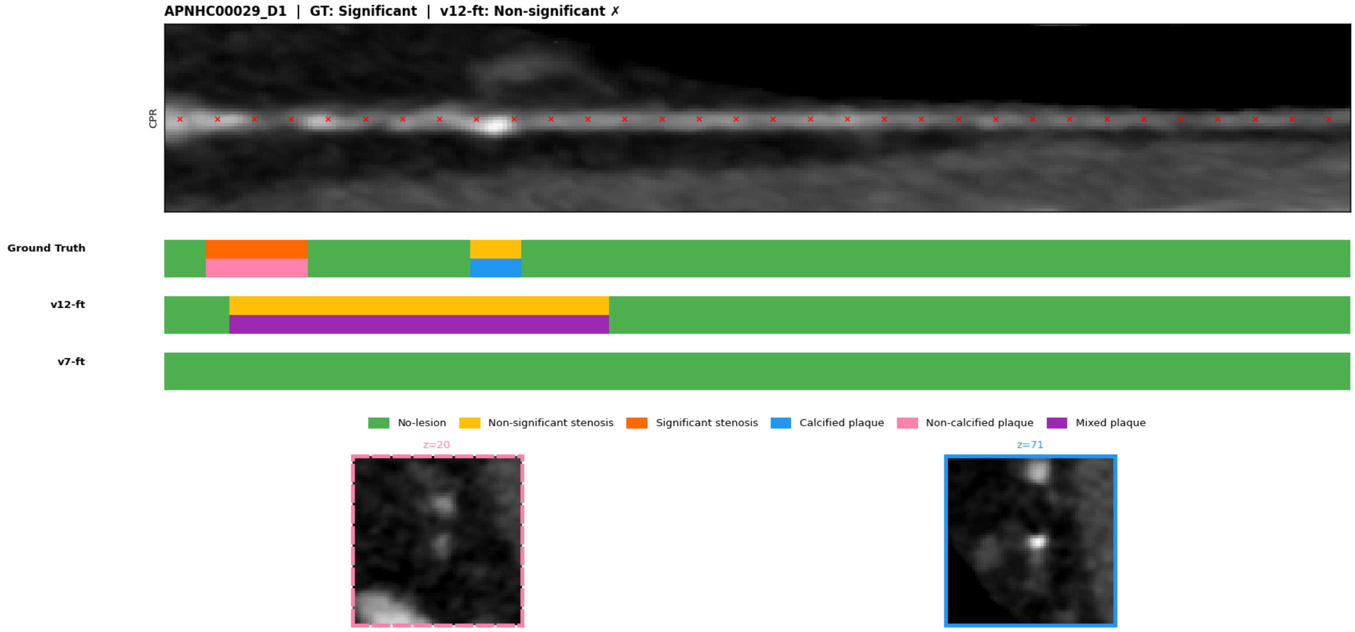
Cases where v7 is correct but v12 is not. These represent the net losses from the v12 changes — important to review.

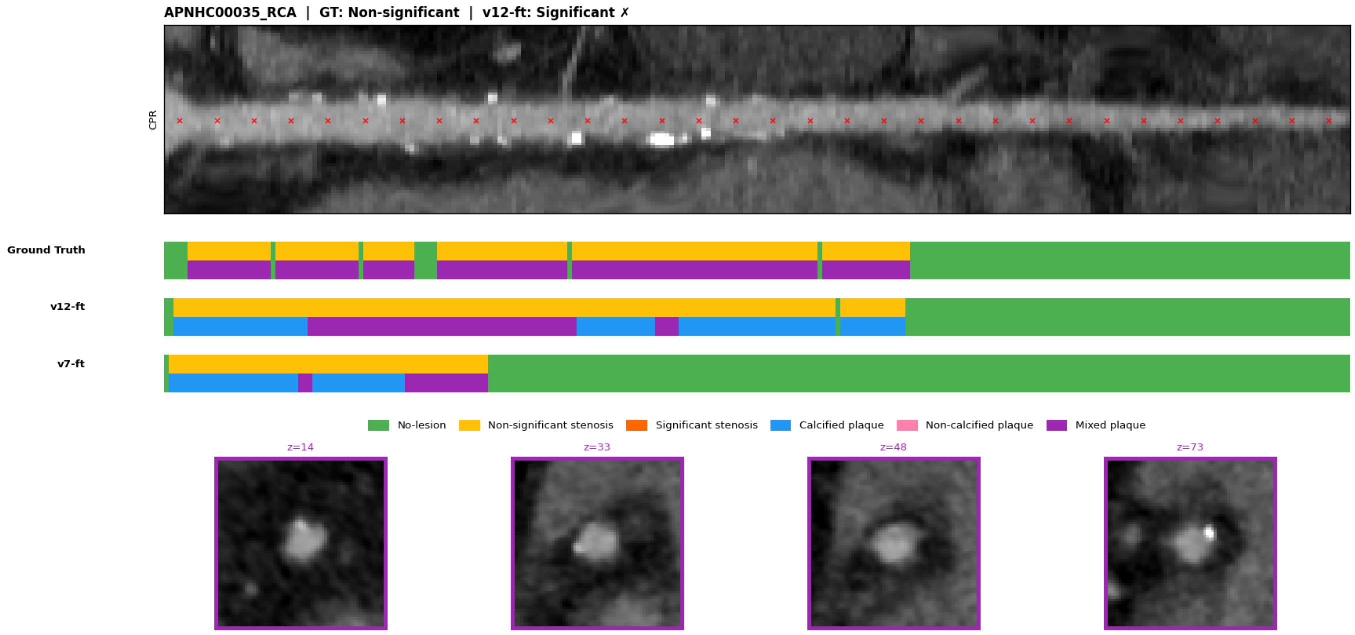




6.4 Both Wrong — Remaining Hard Cases

Cases where neither model gets the right answer. These highlight remaining failure modes.





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7. Key Findings & Clinical Relevance

Non-significant Stenosis Detection Breakthrough

The most clinically significant result in this project is the Non-significant recall going from **0% (v7 standard calibration) → 63.9% (v12 constrained calibration)**. Non-significant lesions are intermediate-severity plaques that do not yet obstruct flow but require monitoring — a model that never predicts this class is clinically unusable despite having high Significant recall.

The constrained calibration framework (`--constrain_nonsig_recall 0.10`) was the key insight: instead of a 2D grid search over `[t_H, t_Sig]` with `t_NS` fixed at 1.0, it performs a full 3D search enforcing `NonSig Recall ≥ 10%`, finding `t_NS=0.65`. This revealed that the model *had* learned Non-significant features — they were simply being suppressed by an overly conservative threshold.

SC Branch Stability

The SC (Sampling Point Classification) branch maintained **ACC=0.814** through v7 → v12, matching the best ever recorded value. This is notable because v9 collapsed the SC branch to 0.322 through a 6× LR mismatch at head re-initialisation. v12's conservative `lr=3e-5` and `T0=60` (ensuring a healthy LR at DC activation) preserved the temporal branch while dramatically improving the spatial OD branch.

Remaining Challenges

- **Non-significant F1 = 0.613** — still the weakest class. The class is inherently hard: subtle wall thickening without flow obstruction, and it sits between Healthy and Significant on a continuous disease spectrum.
- **Mixed plaque F1 = 0.214** — rare class (only 45 samples in test set) with highly variable appearance.
- **Non-calcified plaque F1 = 0.500** — soft tissue plaques are harder to distinguish from vessel wall in CT without iodine contrast enhancement.

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8. Next Steps — v13

v13 fine-tuning is currently running (`logs_finetune_v13.log`). Three architectural improvements over v12:

Improvement	Description	Expected benefit
Parallel 2D/3D streams	Fix: 2D blocks at levels 1+ now receive their own prior output, not 3D output (present in all runs through v12)	Feature diversity in spatial branch
SE attention fusion gate	Per-level learned channel-wise 3D/2D blending (replaces scalar `_3d_weight=0.71`)	Content-adaptive feature fusion
DC temperature annealing	Soft pseudo-labels start at T=3.0, anneal to 1.0 over DC ramp	Stable early fine-tuning with fresh 6-class heads

Expected Stenosis F1: 0.78–0.85

Results will be added to this document once v13 training completes and evaluation runs.
